AQI PM2.5 Demo

Show raw tables

Prediction rows

1000

AQI Big Data (PM2.5) Dashboard

Spark gold layer → Kafka → FastAPI → Streamlit dashboard (predictions)

Model Metrics

	model	rmse	mae	r2
0	linear_regression	3.4354	1.1873	0.9605

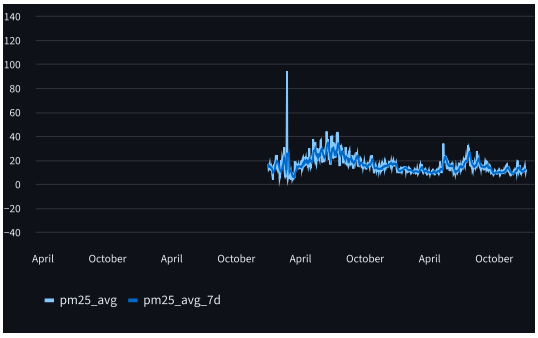
Best Model: linear_regression

R2: 0.9605

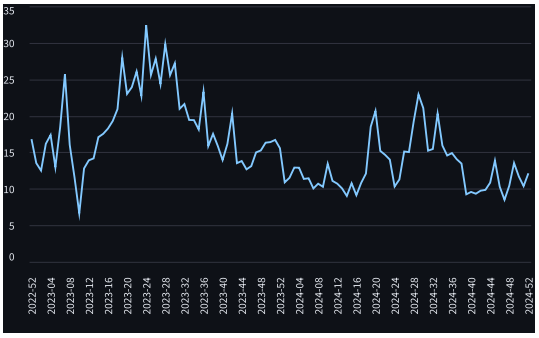
RMSE: 3.44

EDA Trends

Daily Trend (with 7-day moving average)



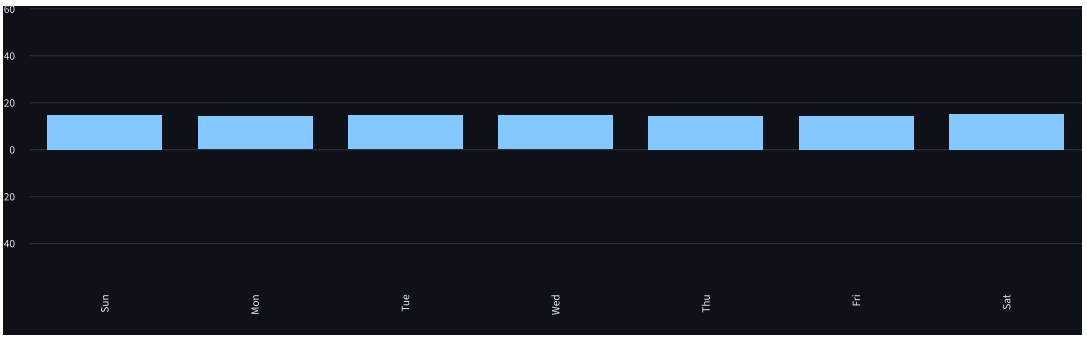
Weekly Trend



	date	pm25_avg	n	pm25_avg_7d
0	2022-12-31 00:00:00	16.8099	276	16.8099
1	2023-01-01 00:00:00	15.5845	857	16.1972
2	2023-01-02 00:00:00	12.1845	826	14.8596
3	2023-01-03 00:00:00	12.7102	868	14.3223
4	2023-01-04 00:00:00	15.144	868	14.4866
5	2023-01-05 00:00:00	16.699	887	14.8554
6	2023-01-06 00:00:00	13.2181	867	14.6215
7	2023-01-07 00:00:00	12.8227	832	14.0519
8	2023-01-08 00:00:00	11.6641	829	13.4918
9	2023-01-09 00:00:00	11.5807	875	13.4055

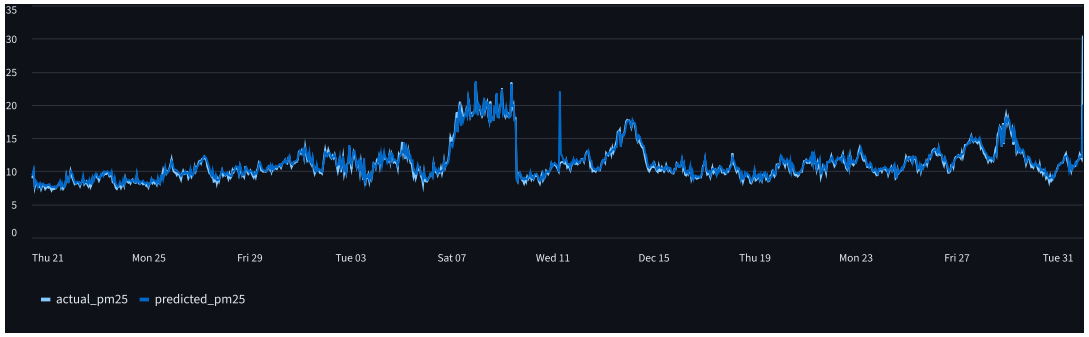
	year	week_of_year	pm25_avg	n	year_week
0	2022	52	16.8099	276	2022-52
1	2023	1	13.5272	5977	2023-01
2	2023	2	12.5053	2365	2023-02
3	2023	3	16.1713	5466	2023-03
4	2023	4	17.379	6716	2023-04
5	2023	5	12.9611	2649	2023-05
6	2023	6	18.4466	229	2023-06
7	2023	7	25.7202	2161	2023-07
8	2023	8	16.0357	263	2023-08
9	2023	9	11.5907	237	2023-09

Weekday Statistics



	day_of_week	n	pm25_avg	pm25_median	pm25_min	pm25_max	day_label
0	1	145586	14.6237	9	0.001	896	Sun
1	2	144321	14.0723	8.8	0.0074	896	Mon
2	3	143671	14.3628	8.916	0.0005	985	Tue
3	4	140245	14.4746	9	0.0005	580.1	Wed
4	5	138352	14.2597	9.0136	0.0025	1001.8	Thu
5	6	141677	14.2337	9	0.0015	985	Fri
6	7	143716	14.9336	9	0.0015	1323.2	Sat

Predictions (Actual vs Predicted)



	date	actual_pm25	predicted_pm25
15432	2024-11-20 09:00:00	9.0276	9.4276
15433	2024-11-20 10:00:00	9.099	9.5203
15434	2024-11-20 11:00:00	9.871	10.1748
15435	2024-11-20 12:00:00	7.5782	7.968
15436	2024-11-20 13:00:00	8.3436	8.5799
15437	2024-11-20 14:00:00	8.1079	8.341
15438	2024-11-20 15:00:00	7.6737	7.5861
15439	2024-11-20 16:00:00	7.6265	8.0687
15440	2024-11-20 17:00:00	7.8987	8.2455
15441	2024-11-20 18:00:00	7.2721	7.616

Trigger the pipeline to refresh data in the backend before demos.